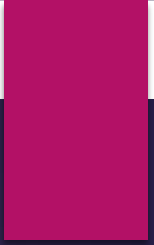


Publishing a Terraform Module to the Terraform Registry

PUBLIC REGISTRY, PRIVATE REGISTRY, VERSIONING & BEST PRACTICES

Learning Objectives

- ▶ Understand Terraform module publishing workflows
- ▶ Differentiate public vs private registries
- ▶ Structure a reusable Terraform module
- ▶ Publish modules to the Terraform Registry
- ▶ Manage module versioning and releases



INTRODUCTION TO TERRAFORM MODULE REGISTRY

What is the Terraform Registry?

Overview

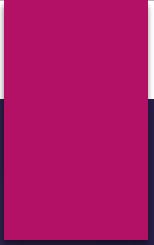
- ▶ Terraform Registry is HashiCorp's official, centralized hub for discovering, publishing, and sharing reusable Infrastructure as Code (IaC) components.
- ▶ It acts as a directory where individuals and teams can find providers, modules , policies, and run tasks designed to simplify Terraform workflows.

Registry Types

- ▶ Public Terraform Registry : anyone can discover, publish, and consume verified or community code.
- ▶ Private Terraform Registry: Internal, organization-owned repositories used to securely distribute proprietary or hardened infrastructure modules

Public vs Private Registry

Public Registry	Private Registry
Open to everyone	Restricted access
Community sharing	Internal organization use
Hosted by HashiCorp	Hosted in Terraform Cloud/Enterprise, GitLab, Bitbucket , Azure DevOps
Public GitHub repos required	Private VCS repositories supported
Ideal for open-source modules	Ideal for enterprise modules



PREPARING A MODULE FOR PUBLICATION

Requirements

- ▶ GitHub Account:
- ▶ HCP Terraform account:
- ▶ Named terraform-<PROVIDER>-<NAME> :
- ▶ Repository description:
- ▶ Standard module structure:
- ▶ x.y.z tags for releases:

Terraform Module Structure

```
tree minimal-module/.
```

```
├─README.md
```

```
├─main.tf
```

```
├─variables.tf
```

```
├─outputs.tf
```

```
└─LICENSE
```

Module Naming Standards

Naming Convention

- ▶ terraform-<PROVIDER>-<MODULE_NAME>

Examples

- ▶ terraform-aws-vpc
- ▶ terraform-aws-ec2

Why Naming Matters

- ▶ Registry discovery
- ▶ Consistency
- ▶ Searchability

Writing Documentation

Essential README Sections

- ▶ Module description
- ▶ Requirements
- ▶ Inputs
- ▶ Outputs
- ▶ Usage examples
- ▶ Providers
- ▶ Version requirements

Documentation Tools

- ▶ terraform-docs

Versioning the Module

Semantic Versioning

- ▶ MAJOR.MINOR.PATCH

Examples

- ▶ v1.0.0
- ▶ v1.1.0
- ▶ v2.0.0

Importance of Versioning

- ▶ Stability
- ▶ Compatibility
- ▶ Safe upgrade

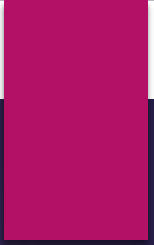
Git Tags & Releases

Creating a Release

- ▶ `git tag v1.0.0`
`git push origin v1.0.0`

Registry Uses Tags For

- ▶ Module versions
- ▶ Release tracking



PUBLISHING TO THE PUBLIC TERRAFORM REGISTRY

Publishing Workflow

1. Create a GitHub repository for your module in the following format – terraform-<Provider_Name>-<Module_Name>
2. Create your Terraform module configuration
3. Push the content of your module configuration to the created repository.
4. Create a tag for your repository.
5. Login to Terraform Registry
6. Connect GitHub account
7. Click on publish and select module.
8. Select that GitHub repository.
9. Success, your module has been published!